Catch Me If You Can: Quantifying Outfield Shifting Impact for Baseball Fans

Team 150

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Abstract

Defense is the black sheep of baseball, overlooked and underappreciated. There is a lot that goes into making a catch in the outfield, and most fans only recognize the last few seconds of a play as to whether it was a catch or an error. However, there are many cases where a spectacular play was made, but it looked like a routine catch. A lot of plays are properly defended before the batter even makes contact with the pitch. This project calculates catch probability on balls hit to the outfield compared to the catch probability of the same ball where the outfielder would start from at the average position at contact. Using the SMT-provided player and ball positioning data, calculate the optimal route distance, hang time, attack angle (the angle at which the fielder needs to turn to field the ball), and the descent angle of the ball to create a catch probability system that can be applied to both actual results of the play, and a what-if scenario where the player had not shifted in preparation for the batter. This allows for a catch-probability based presentation that can help the fans understand the impact of outfield defensive shifting.